

TYPES OF CHEMICAL REACTIONS

Teacher Lesson Guide

1. Metadata

Topic	Types of Chemical Reactions
Level	Class X
Lesson Length	60 minutes

2. Learning Objectives

- LO1: Identify and describe combination reactions, noting that they are often exothermic, and write balanced equations for them.
- LO2: Distinguish decomposition reactions from other types and construct balanced equations for given examples.
- LO3: Explain single displacement, double displacement, and precipitation reactions with textbook examples.
- LO4: Identify oxidised and reduced substances in redox reactions and describe the underlying electron transfer.
- LO5: Compare exothermic and endothermic reactions, linking the concepts to the reaction types studied.

3. Prerequisite Knowledge

- Atoms are conserved during chemical reactions; no atoms are created or destroyed.
- A chemical equation must be balanced to reflect conservation of atoms.
- Basic symbols for states of matter (s, l, g, aq).
- Understanding of heat energy as a form of energy transferred during reactions.

4. Key Terms

Term	Definition
Combination reaction	Two or more substances combine to form a single new substance.
Decomposition reaction	A single substance breaks down into two or more different substances.
Single displacement reaction	An element displaces another element from its compound.
Double displacement reaction	Two compounds exchange ions to form two new compounds.
Precipitation reaction	A type of double displacement reaction that produces an insoluble solid (precipitate).
Oxidation	Gain of oxygen or loss of hydrogen in a substance during a reaction.
Reduction	Loss of oxygen or gain of hydrogen in a substance during a reaction.
Exothermic reaction	A reaction that releases heat to the surroundings.
Endothermic reaction	A reaction that absorbs heat from the surroundings.

Term	Definition
Balanced chemical equation	An equation in which the number of atoms of each element is the same on both sides.

5. Key Points

- When two or more substances combine to form a single product, the reaction is a combination reaction.
- Many combination reactions release heat, making them exothermic.
- A balanced equation has the same number of atoms of each element on both sides, reflecting conservation of mass.
- In redox reactions one reactant is oxidised and the other is reduced.
- When ions exchange between two compounds, a solid precipitate may form.
- The textbook excerpt does not provide specific decomposition examples, but the type involves a single compound breaking into two or more simpler substances.
- The provided pages do not detail single displacement reactions; they involve an element replacing another in a compound.

6. Real-World Applications

Energy from Respiration

Respiration in the human body is an exothermic process where glucose reacts with oxygen to release energy, carbon dioxide, and water.

Think about: Why does the body feel warm after a meal, based on the chemical nature of respiration?

Expected answer: Because the oxidation of glucose with oxygen releases heat (exothermic), raising body temperature.

Suggested time: 4 minutes

Combustion of Natural Gas for Home Heating

Burning natural gas (CH_4) in furnaces provides heat for homes; the reaction is exothermic, converting chemical energy to thermal energy.

Think about: How does the exothermic nature of the CH_4 combustion reaction make it useful for heating?

Expected answer: The reaction releases a large amount of heat, which can be transferred to the surrounding air to raise indoor temperature.

Suggested time: 3 minutes

7. Teacher Schedule

Time	Stage	Teacher Direction	Student Action
0-5 min	Introduction	Introduce the lesson topic and learning objectives. Use brief retrieval questioning to activate relevant prerequisite knowledge.	Respond to retrieval questions and review the learning objectives.

Time	Stage	Teacher Direction	Student Action
5-14 min	Teaching Phase 1	Emphasise the formation of a single product and heat evolution (e.g., $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{heat}$). Use the real-world application 'Energy from Respiration': Respiration in the human body is an exothermic process where glucose reacts with oxygen to release energy, carbon dioxide, and water. Ask students: Why does the body feel warm after a meal, based on the chemical nature of respiration? Use the real-world application 'Combustion of Natural Gas for Home Heating': Burning natural gas (CH_4) in furnaces provides heat for homes; the reaction is exothermic, converting chemical energy to thermal energy. Ask students: How does the exothermic nature of the CH_4 combustion reaction make it useful for heating?	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
14-23 min	Teaching Phase 2	Highlight that a single reactant breaks into two or more products (e.g., $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$).	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
23-34 min	Teaching Phase 3	Show how an element or ion replaces another and how an insoluble salt may form.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
34-43 min	Teaching Phase 4	Use the $\text{Na} + \text{O}_2$ example to illustrate loss/gain of electrons.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
43-50 min	Teaching Phase 5	Connect heat release/absorption to observable temperature changes.	Listen, annotate notes, respond to checks for understanding and apply knowledge to examples.
50-55 min	Question Phase	Students complete Q1, Q2, Q3, Q4, Q5. Review answers and address remaining gaps in understanding.	Answer the lesson questions and review responses.
55-60 min	Reflection / Review	Review the learning objectives and identify areas requiring further review.	Review the learning objectives and identify any topic requiring further review.

8. Activities

Suggested Activities If Time Available

Write and Balance Combination Equations (12 min)

Students choose two substances (e.g., carbon and oxygen, hydrogen and oxygen) and write the correct chemical equation. Then they balance the equation, checking atom counts on both sides.

Identify Oxidised and Reduced Species (13 min)

Using the redox examples $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ and $\text{MnO} + 2\text{HCl} \rightarrow \text{MnCl}_2 + \text{H}_2\text{O} + \text{Cl}_2$, students label which reactant is oxidised and which is reduced, and write the half-reactions.

9. Questions and Answers

Q1. What is a combination reaction? Give an example from the textbook.

Answer: A reaction where two or more substances combine to form a single product; e.g., $\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{heat}$.

Q2. Write the balanced chemical equation for the decomposition of zinc carbonate as presented in the source.

Answer: $\text{ZnCO}_3\text{(s)} \rightarrow \text{ZnO(s)} + \text{CO}_2\text{(g)}$

Q3. When potassium bromide reacts with barium iodide, what type of reaction occurs and what are the products?

Answer: It is a double displacement (precipitation) reaction; products are $\text{K}^+\text{I}^- \text{(aq)}$ and $\text{BaBr}_2 \text{(s)}$.

Q4. In the reaction $4\text{Na(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{Na}_2\text{O(s)}$, which substance is oxidised and which is reduced?

Answer: Na is oxidised (loses electrons) and O_2 is reduced (gains electrons).

Q5. Explain why the reaction of calcium oxide with water is described as exothermic.

Answer: The reaction releases a large amount of heat, causing the temperature of the mixture to rise, which matches the definition of an exothermic reaction.